

The AI job loss story is all about bundles

Evidence on the patterns of white-collar work displacement is beginning to match the theory, write John Burn-Murdoch and Madhumita Murgia

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For this edition we are revisiting the big question of whether AI has already started to take white-collar jobs, in light of a flurry of new research and evidence. Sarah is away this week, so stepping into her shoes is the FT's AI editor, Madhumita Murgia.



Today's edition was sparked by a new paper from economists Leland Crane and Paul Soto of the US Federal Reserve, which represents the first time to my knowledge that official labour market statistics have corroborated the story from detailed private payroll data that AI is reducing employment in some pockets of the economy.

We have previously reported on research showing a dip in employment for young software developers, based on fine-grained analysis of millions of payroll records, but the finding wasn't matched by labour force survey data. That gap has now closed.

By using an expanded definition of coders — crucially including the large contingent of contractor software developers as well as coders outside the tech industry — Crane and Soto find a very similar result using the flagship US labour force survey as Stanford's Erik Brynjolfsson and co-authors did using payrolls, with both estimating that around half a million fewer coders are working today than would have been had pre-LLM-era employment trends continued. It's worth noting that this is not an absolute decline in coder employment, but a marked slowdown in growth.

Just as interesting as the headline findings is the fact that nuances in these results align neatly with several recent papers setting out frameworks for thinking about how AI job displacement is likely to play out and highlighting weaknesses with simple occupation-based or task-based models.

A paper last month by LSE professor Luis Garicano and co-authors extends the idea that jobs are bundles of tasks, to consider whether the different activities in a job are a tightly

bound bundle or something more akin to an itemised list of discrete activities. In the context of software, a contractor or junior hire generally falls into the latter group: these jobs are weak bundles, with daily work consisting mainly of writing code to spec — tasks that could be given to someone else (or AI) without any disruption to the workflow or the quality of the final product. Here, AI breaks off a large chunk of the job and leaves a role with substantially diminished scope (or obviates the need for that hire or contract).

But senior developers, or coders working outside the tech industry in roles where their programming skills are combined with domain-specific expertise, tend to have jobs comprising tightly enmeshed and cross-functional tasks. Here extracting the coding part of the job from all the rest is much harder, so the bundle of tasks remains intact. Instead of becoming a competitor AI becomes an assistant, enhancing rather than eroding the job. This fits with the findings from Brynjolfsson and our own analysis that hiring for senior software roles continues to hold up better than for junior ones.

The bundling framework is also explored in recent papers by Lukas Freund and Lukas Mann, and by Joshua Gans and Avi Goldfarb, who move beyond the size and interconnectedness of a job's task bundle to consider the importance of the surviving tasks left after one is automated. When coding is done by AI, senior developers have more time to spend on the many other valuable parts of their job, like translating business needs into product specifications or making judgment calls based on years of accumulated expertise.

AI automates a relatively lower-value part of their job and acts as a multiplier on all the rest. But take away coding from a junior developer or contractor and you're left with very little. In this way, the same technological capability shrinks one job while expanding another — moreover it erodes the junior version of a job even as it enhances the senior version.

Between the now-consistent picture on junior coding employment and the expanded framework of jobs as bundles of tasks, it feels to me like we're developing an increasingly coherent picture of AI job displacement.

This new data comes at an interesting moment, John. OpenAI released a policy blueprint this week that proposes some radical changes to the social contract, in response to what it casts as inevitable job losses and disruption of entire sectors. Of course, it's in their interest to claim their product will be singularly revolutionary, but I've also spent the past couple of weeks speaking to investors, analysts and executives from a range of white-collar professions on which jobs are resilient in an age of AI agents. The fact that AI is shrinking certain types of work — mainly early-career jobs — is accepted in these circles, although it is being whispered.

Taken together, the recent research is really interesting for two reasons. First, it confirms what anecdotally AI companies, white collar professionals and pretty much any AI user I speak to, are telling me: that AI automation is a double-edged sword.

On one hand, AI allows you to supercharge your skills if you are already proficient at your job. One person from a frontier AI company described this as tackling a gnarly project by cloning yourself. On the other hand, if you are just starting out, and don't yet have the instincts and knowledge developed through hands-on experience, you are more likely to be replaceable.

I find it fascinating that this effect seems to be profession-agnostic. I've heard this repeated from software engineers, but also journalists, musicians, financial services professionals and lawyers. This is partly because of how the technology inherently works: the errors it makes can often seem random, meaning those without the nous to doubt its outputs and sense-check AI-assisted work are caught out by mistakes more easily. It seems using AI effectively has an art to it — and that skill only comes with mastery of your subject.

The other thing the research points to is that AI is picking off clusters of tasks that make up a job, starting with the mundane and working its way up the chain. The wave of AI agents that we see today, like Anthropic's Claude Cowork or even OpenAI's Codex, which use AI to write code, can complete multiple tasks simultaneously, extending the complexity of what the technology can accomplish independently.

On these topics, I recently sat down with Mark Chen, head of research at OpenAI, who pointed me to the METR benchmark, a metric that measures AI performance in terms of the length of tasks AI agents can complete.

This horizon has been expanding rapidly, with an exponential increase over the past months. He told me that a year ago, they were dealing with AI completing tasks that humans could do in minutes. “Now we're dealing with tasks in the hours. And if you extrapolate that forward, we're soon going to be having our models do tasks reliably that would take humans days,” he said, referring to building a self-contained, functional piece of software, for example.

It struck me that although job displacement currently seems to be impacting junior employees disproportionately, that won't be the case for much longer. The advent of AI agents that can handle higher-level tasks means it is a matter of time before even experienced workers get pushed into being AI project managers, rather than creative thinkers. Do you have a more hopeful outlook, John?

Thanks Madhu, it's really interesting to hear those industry whispers matching the hard data. There's certainly no sign that the forward march of AI's capabilities is slowing, and

that will mean more disruption of jobs. But there are two parts of this where I'm more optimistic.

The first is that I think there's a reason we're still seeing relatively little evidence of job displacement outside of coding (more on this next week) and that this may hold true for some time.

And the second is that as someone who has increasingly become a manager of agentic AI projects, I've found it great fun, and more a multiplier of my creativity than a replacement of it.